AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 24 · STUDENT LESSON

Train, Validation, Test

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Fit on training. Choose on validation. Report the frozen test.

Assign data roles and grouped source membership to the actual decisions they support.

Keep one habit: Complete candidate results → pictured selection tie → twenty action assignments → grouped90-image audit → independent new results.

MAIN LESSON · SECTION 1

Which job is this dataset doing?

> Assign fitting,development selection and final reporting to their actual information roles.

Learning goals

I can select a model using a declared **validation** rule.

I can keep **training**, **validation** and test roles separate.

I can recognize repeated **validation** as development.

Recall final-test exposure

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.
- Which decisions may development validation results guide,and why can those same inspected results no longer be called an untouched final test?

End goal

By the end, I can explain train, **validation**, test using a traced example and a stated limitation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Training fits parameters, **validation** chooses settings, and testing estimates final performance after choices are frozen.
</details>



LEARNING TARGET You have 30 leaves with 3 views each. Allocate 18 leaves to train,6 to **validation**,6 to test. How many images are in each? Where is a learned normalisation mean fitted?

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Separate fitting a candidate from choosing among candidates

> Keep related source views together before assigning roles.

Complete role definitions and fictional scope

Fitting and selecting answer different questions. **Training** adjusts parameters. **Validation** helps choose settings or model families. A final test estimates the performance of the frozen development **process** on held-out examples.

Keep related records in the same split when separate rows could reveal the same source.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **validation** and model selection.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.
</details>



MYSTERY The team must use train, **validation**, test to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Apply the declared validation-first selection rule

> A training-perfect candidate can lose on development evidence.

Complete candidate result table

Candidate	Supplied training MAE cm	Supplied validation MAE cm	Declared selection fact
A	1	3	Simpler than C under the declared tie policy
B	0	5	Perfect training score does not decide selection
C	2	3	Ties A on validation, loses declared simplicity tie

Trace the supplied choice and its rule

Three candidates have **training** MAE A=1, B=0, C=2 and **validation** MAE A=3, B=5, C=3. The declared rule is smallest **validation** MAE, then simpler model, with A simpler than C. Choose A.

B fits **training** perfectly but does worse on **validation**. The final test remains unopened.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

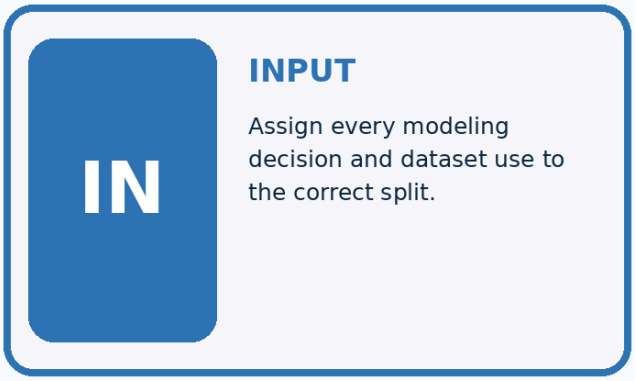
Keep the distinction between an observed result and a proposed next step.

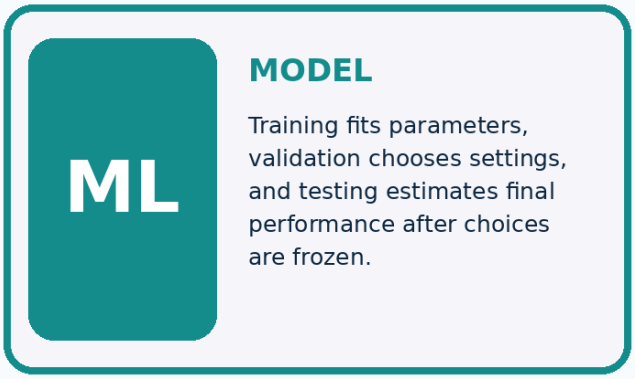
▼ **Earlier wording, alternatives and picture transcripts**

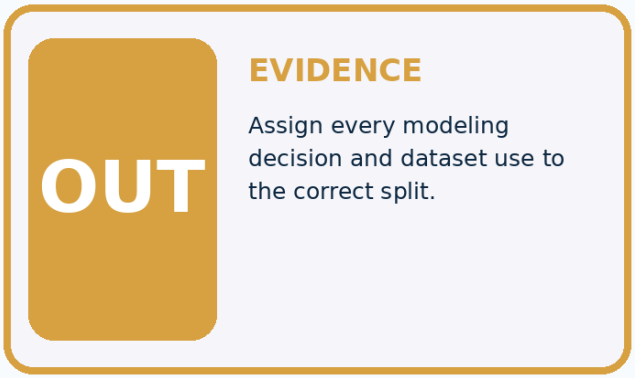
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Assign every modeling decision and dataset use to the correct split.

MODEL IDEA **Training** fits parameters, **validation** chooses settings, and testing estimates final performance after choices are frozen.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Fit two candidate trees on **training** rows. Choose depth and threshold using **validation** metrics and error slices.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

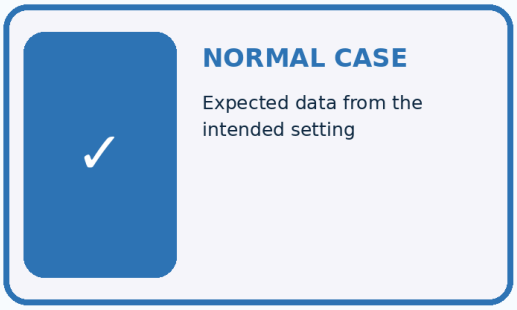
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

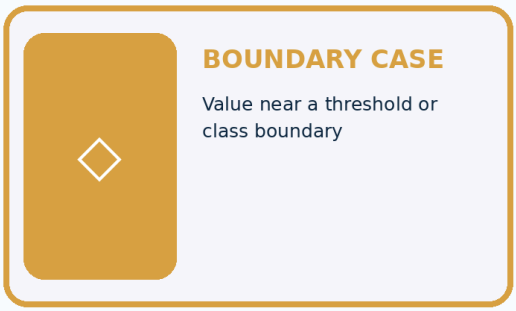
Compare candidate fitting with validation selection

> The final test does not resolve a development tie.

Four evaluation conditions



NORMAL Expected data and expected behavior.

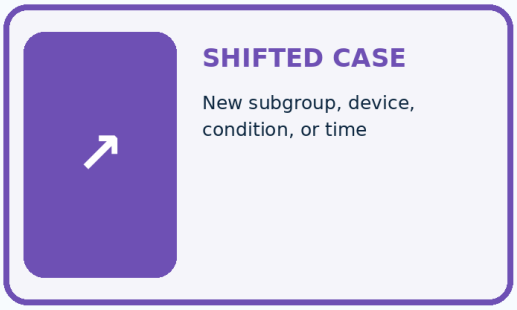


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule.

▼ **Check after predicting**

Prediction: Select A using the predeclared simpler-model tie-break.

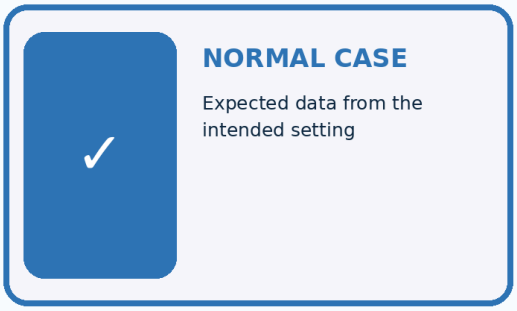
Reason: The **validation** errors tie, so the stated secondary rule decides without consulting final data.

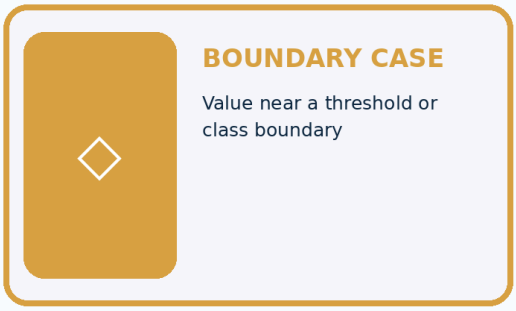
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

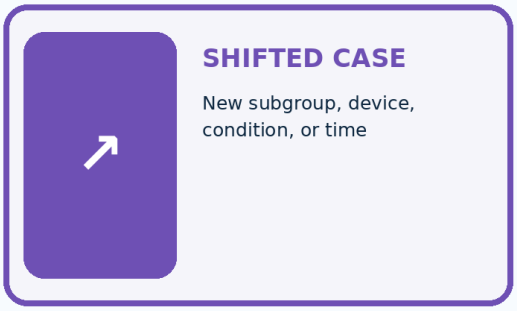
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read the candidate’s role before its lowest number

> A recorded tie-break avoids consulting final outcomes to select a winner.

Complete pictured facts and separate-result scope

Assign a separate job to **training**, **validation** and final test **evidence**.

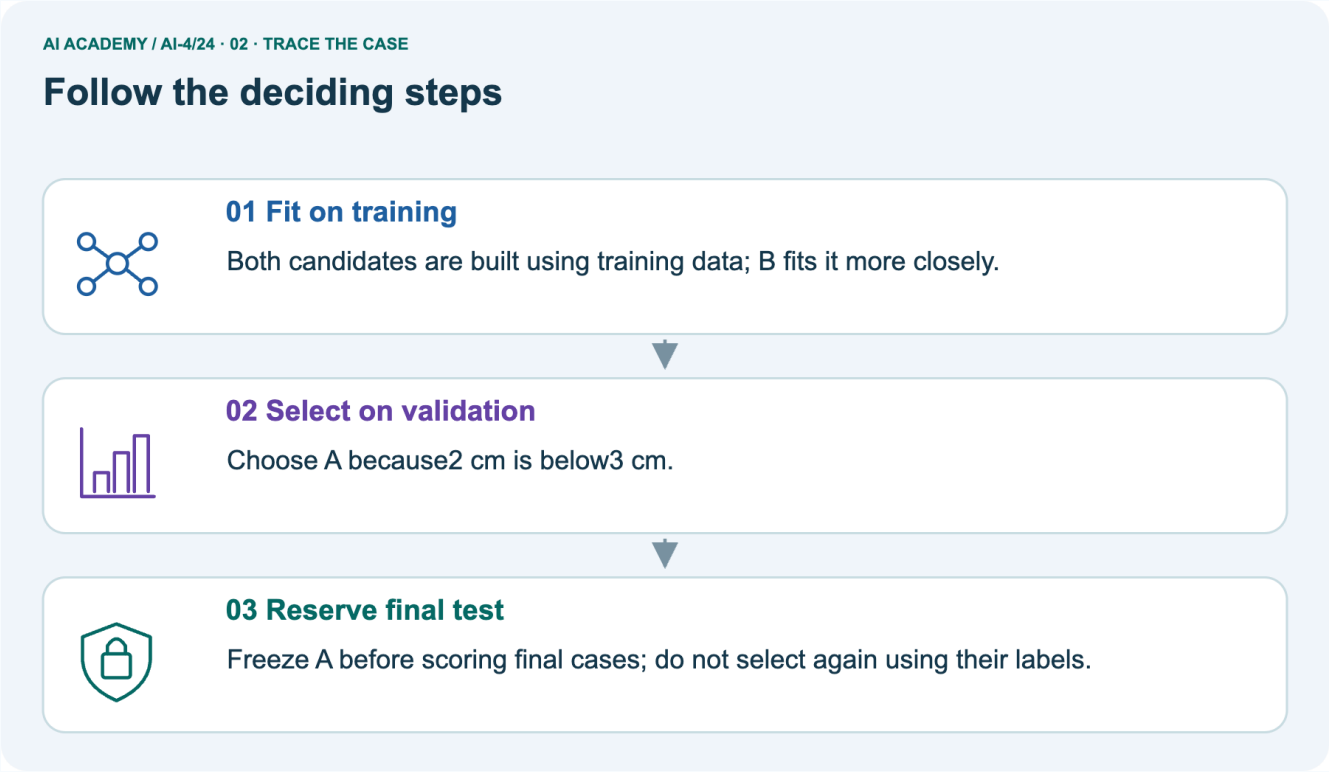
Candidates A and B have **training** MAE1 and 0 cm; **validation** MAE2 and 3 cm. **Selection rule** is smallest **validation** MAE.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main A/B/Cresults choose A at validation MAE 3 cm, tied with C but simpler; Bhas training 0 and validation 5. Separate picture compares only A/Bwith validation 2/3 cm, again choosing A. The pictured change ties both at2 cm; predeclared simplicity still chooses A. These are supplied development results, not training code or final-test scores.

A lower inspected validation number does not guarantee population performance, and repeated choices can adapt to the validation sample.

Trace fitting,selection and reserved final check



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fit on training	Both candidates are built using training data; B fits it more closely.
2. Select on validation	Choose A because2 cm is below3 cm.
3. Reserve final test	Freeze A before scoring final cases; do not select again using their labels.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

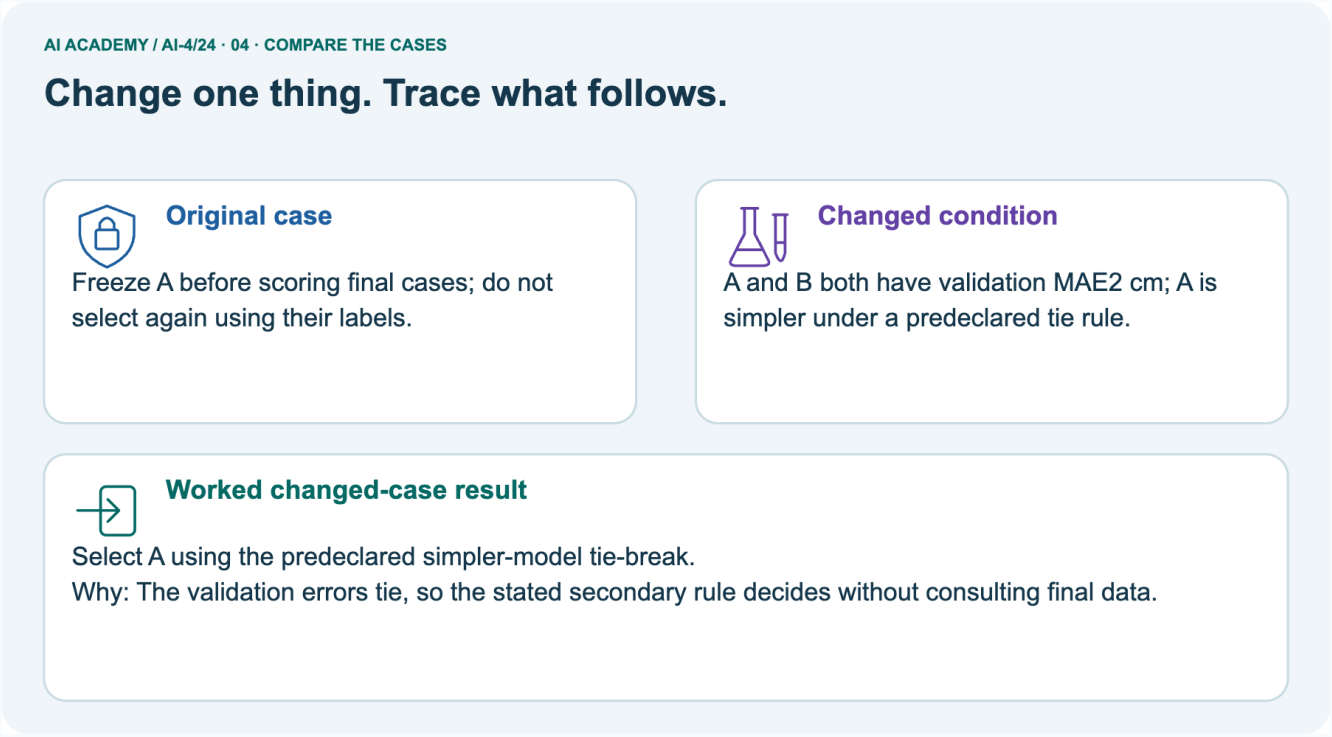
Read the picture in order. Freeze A before scoring final cases; do not select again using their labels. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict the pictured validation tie

- Change one thing: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain exposure and selection limits

- Repeated **validation** use can itself overfit model choices; keep a genuinely untouched final evaluation.

- Read the labelled picture: Why does B's zero **training** MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the three data-role questions

Validation and model selection at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Training	Fit each candidate	No test labels
Validation	Choose A by rule	Record every attempt
Final test	Frozen model only	Report after selection

A tie policy prevents a convenient choice after inspecting results. Equal validation scores do not prove equal future performance. Record candidate settings, split identifiers, preprocessing and the full comparison, including rejected options.

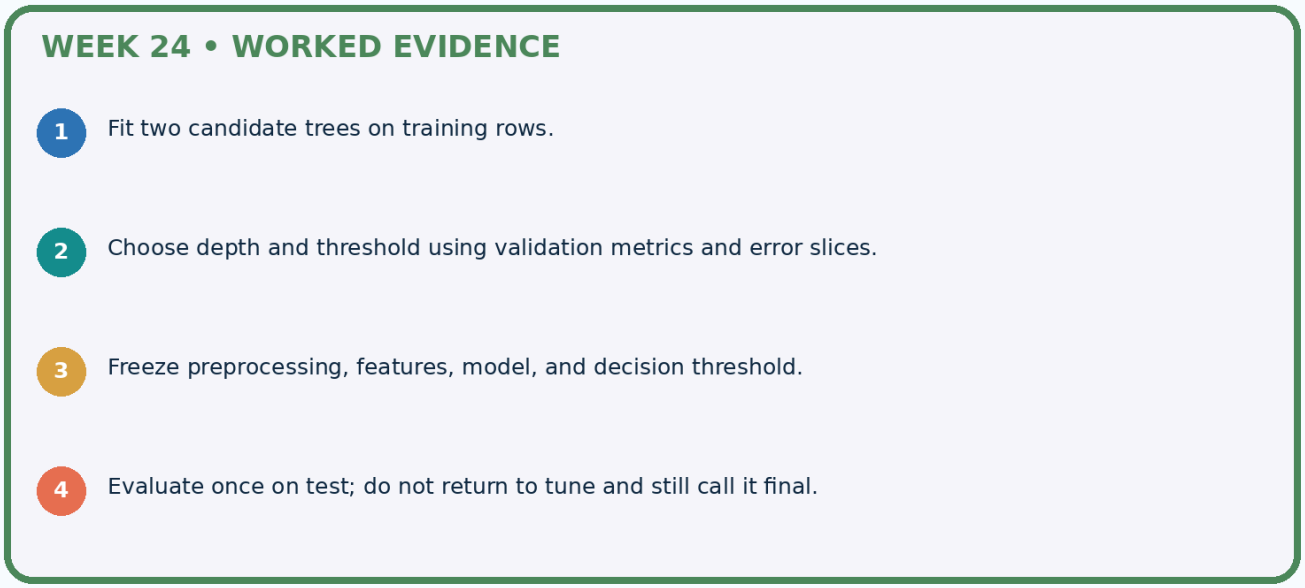
Where should parameter fitting, setting selection and final reporting occur? Explain the steps or evidence behind your response.

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.



STEP 1 Fit two candidate trees on **training** rows.

STEP 2 Choose depth and threshold using **validation** metrics and error slices.

STEP 3 Freeze pre**process**ing, features, model, and decision threshold.

STEP 4 Evaluate once on test; do not return to tune and still call it final.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

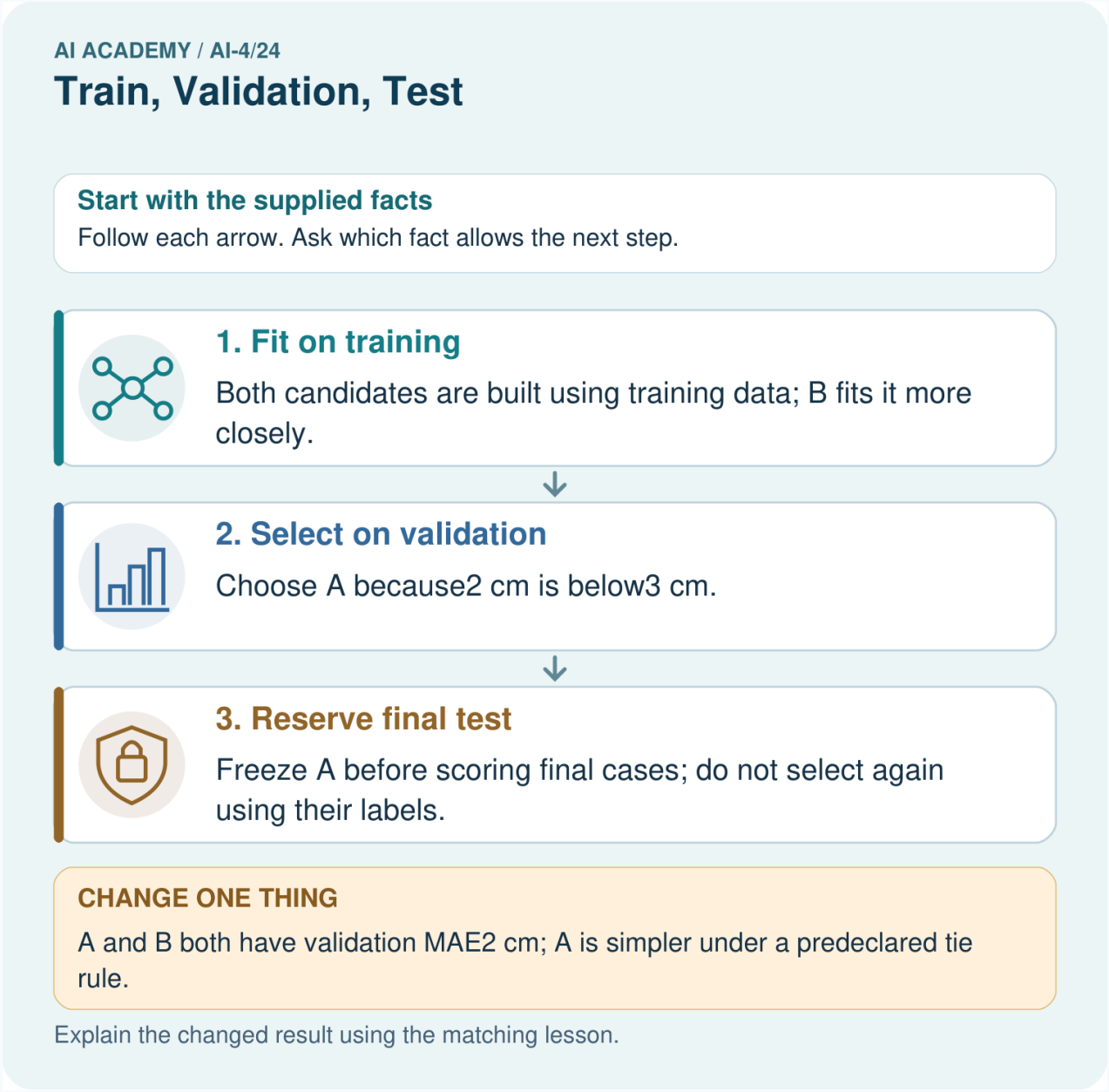
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Train, Validation, Test

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the test used as a development tie-break

> The role follows how information influenced choices.

Diagnose the claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE Three random folders are enough even when photos of one plant appear in each.

Explain limits and repair

- A low score is meaningful only with its unit, metric, evaluated population and baseline. Decide how to handle ties and missing **prediction**s before comparison.

  If no appropriate final test exists, state that limitation and keep claims about **validation**.

> **Why is the winning validation score not a guaranteed future score?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Three random folders are enough even when photos of one plant appear in each. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

MAIN LESSON · SECTION 8

Practise roles and record every development attempt

> Adding a candidate after inspecting validation errors is another logged development choice.

Try supported W2

Use workbook W2 for the connected example “**Validation** and model selection”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Trace a new candidate and repeated validation use

If you inspect **validation** errors and invent candidate D, that is another development attempt. Log it. Repeated experiments can adapt to peculiarities of **validation**. Do not rename **validation** as an untouched test.

More rows alone do not repair duplicated groups or future information leaking into **input**s.

Explain the evidence beyond a final choice

- Use the worked case for **Validation** and model selection. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain why training-zero does not choose pictured B

- Why does B’s zero **training** MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Reserve the fresh P/Q/R independent task

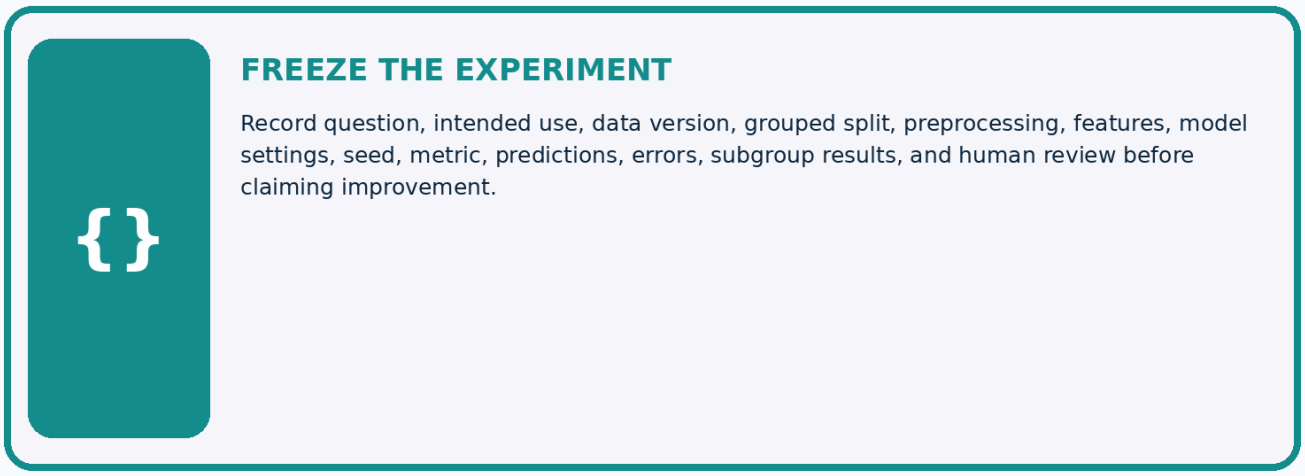
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Assign data roles

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run all twenty role assignments and audit a complete grouped split

> The source supplies every action and every leaf/view range needed for the investigation.

Twenty complete actions to classify

Action ID	Complete action to assign or repair
A01	Fit a numeric mean baseline using training targets
A02	Learn min/max scaling from training feature rows
A03	Learn a decision-tree root and leaf labels from training
A04	Store labelled training examples for a neighbour method
A05	Select a candidate slope/intercept by training SSE
A06	Choose the majority-label constant from training labels
A07	Compare predeclared k values on validation examples
A08	Choose tree depth using validation predictions
A09	Choose between model families using the declared validation metric
A10	Compare proposed feature sets with validation evidence

Action ID	Complete action to assign or repair
A11	Choose a decision threshold using a declared development policy
A12	Apply the predeclared simpler-model tie rule to equal validation scores
A13	Log a new candidate invented after inspecting validation errors
A14	Record the frozen model, preprocessing, threshold, metrics and membership before final-label inspection
A15	Run the fixed chosen system on the reserved final cases
A16	Compute final metrics using all planned cases and the declared failure policy
A17	Publish final counts, collection limits and the exact model/data version
A18	Tune a threshold after reading test errors, then claim a fresh final score on the same rows
A19	Fit normalization on training plus validation/test rows before splitting
A20	Randomly split different views of the same plant across sets for an unseen-plant claim

▼ Check roles after trying

Action ID	Role / repair judgment
A01	Train:fit
A02	Train:fit
A03	Train:fit
A04	Train:fit
A05	Train:fit
A06	Train:fit

Action ID	Role / repair judgment
A07	Validation:select
A08	Validation:select
A09	Validation:select
A10	Validation:select
A11	Validation:select
A12	Validation:select
A13	Validation:development record
A14	Final-test preparation
A15	Test:evaluate
A16	Test:evaluate
A17	Test:report
A18	Invalid final claim; post-test development
A19	Invalid training-only preprocessing boundary
A20	Invalid source grouping for that claim

Complete grouped leaf/view assignments

Complete fictional leaf/view manifest	Assigned role	Leaf count / image count
L01–L18, each with views a, b, c; all three views stay with its leaf	Training	18 leaves /54 images
L19–L24, each with views a, b, c	Validation	6 leaves /18 images
L25–L30, each with views a, b, c	Reserved final test	6 leaves /18 images

Complete fictional leaf/view manifest	Assigned role	Leaf count / image count
Total source and audit	L01–L30 with3 views each, no omitted or duplicated IDs	30 leaves /90 images

Normal,tie-policy,missing-source and new-candidate probes

Complete role-assignment probe	Expected bounded check
Normal declared A/B/Cvalidation results	AandCtie at3 cm; choose A by declared simplicity, not Btraining0.
Boundary equal validation MAEwithout a predeclared tie policy	Declare/document a development tie policy without consulting final results; do not pretend it was predeclared if it was not.
Missing plant IDfor extra image X	Investigate source identity or document exclusion before claiming grouped independence.
Changed candidate Dafter inspecting validation errors	Log as development; the final set remains protected until choices are frozen.

Assign,audit and record exposure

1. Role Assignment: assign all twenty actions to the actual fitting, development or final-check stage, and mark the invalid workflows with a repair. Then audit the complete 30-leaf/90-image manifest:each view once, each leaf one role, no group overlap. Explain why an image count is not an independent-leaf count and why learned normalization uses training only.

   The18/6/6 allocation is a classroom example, not a universal split percentage. Candidate metrics are supplied, no candidate training program or final score is invented. Keep the selected P/Q/Rresults for an uncoached first attempt.

Keep evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

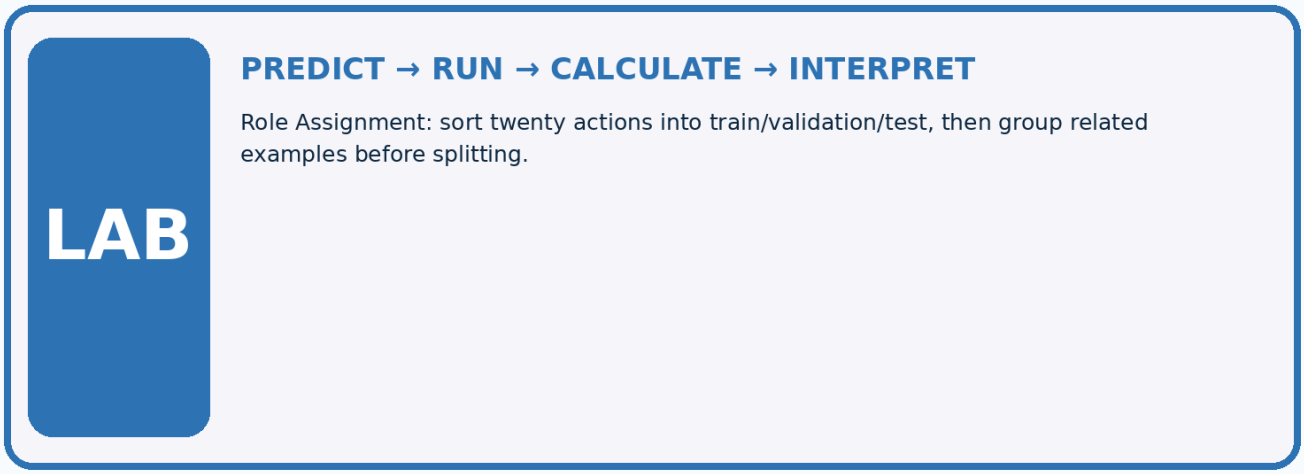
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Role Assignment: sort twenty actions into train/**validation**/test, then group related examples before splitting.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: choose from new development results

> Name the selection evidence and protect the separate final check.

Complete independent result table in the task and scope

Training errors P 2 Q 0 R 1; **validation** errors P 4 Q 6 R 4. Rule: smallest **validation** MAE, then simpler; P is simpler than R. Choose and state the test action.

This selected task uses fresh P/Q/Rtraining and validation MAEs, with Pdeclared simpler than R. Apply the stated validation-first rule and tie-break; do not substitute the main A/B/Cnumbers. No final-test values are supplied, and none are needed to select the candidate or state the freeze-and-test action.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply independently

Week 24 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Assign roles: set A fits model parameters; B selects between two models; C is opened once after choices are fixed. Which is train, **validation** and test?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Publish the grouped **split strategy** and counts for the Plant Classifier, including why plant identity stays within one split.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

You have 30 leaves with 3 views each. Allocate 18 leaves to train,6 to **validation**,6 to test. How many images are in each? Where is a learned normalisation mean fitted?

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's split strategy tied to its intended claim

> A role table should explain both the source grouping and each decision it supports.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Publish the grouped **split strategy** and counts for the Plant Classifier, including why plant identity stays within one split.

Save source/group IDs and exact role assignments, both source and image counts, the intended prediction claim, training-fitted preprocessing, and all development attempts. Record selection metrics, tie policy and freeze version before final outcomes. Do not describe unknown source identity or repeated validation use as untouched independence.

Keep the classroom allocation conditional on the available groups and intended-use evidence.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits,uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

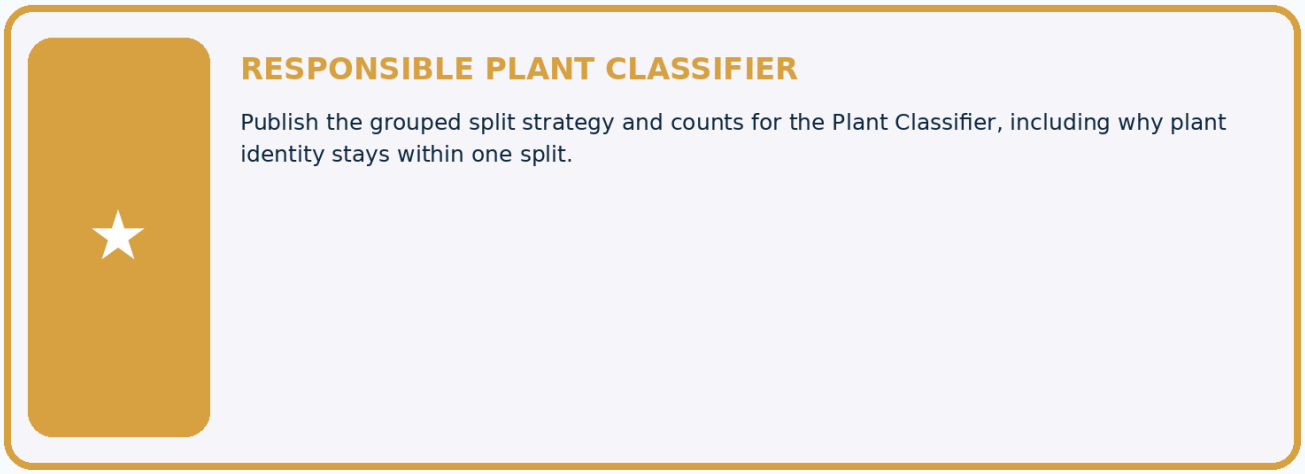
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish by matching each decision to its evidence role

> Explain why a folder label or a perfect training score cannot substitute for that record.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Select a model using a declared **validation** rule

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
training set	data used to fit parameters	
validation set	data used to choose settings	
test set	data used for final estimate	
split strategy	rule assigning examples to sets	

Closer-explanation terms

Word	Meaning
Validation	Development evidence used to choose between options
Hyperparameter	A model setting chosen outside its fitted parameters
Selection rule	A declared method for choosing an option

Word	Meaning
Test set	Held-out evidence reserved for a final evaluation

▼ Earlier wording, alternatives and picture transcripts

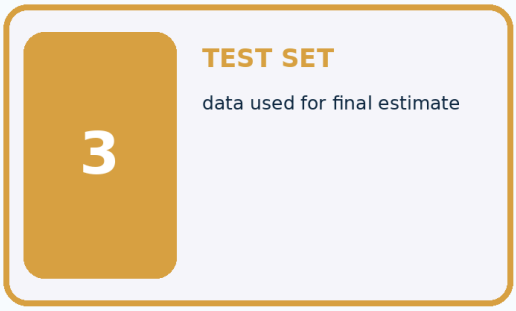
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR Three random folders are enough even when photos of one plant appear in each.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **training set**, **validation set**, **test set**, **split strategy**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE Three random folders are enough even when photos of one plant appear in each.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

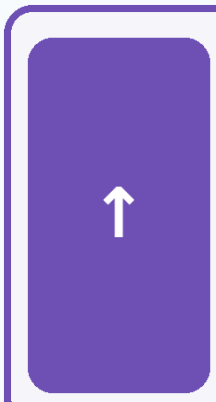
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



ML BUILDER MASTERY

1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

▼ **Reference · Inspect your evidence**

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh P/Q/Rvalidation-and-test-action task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

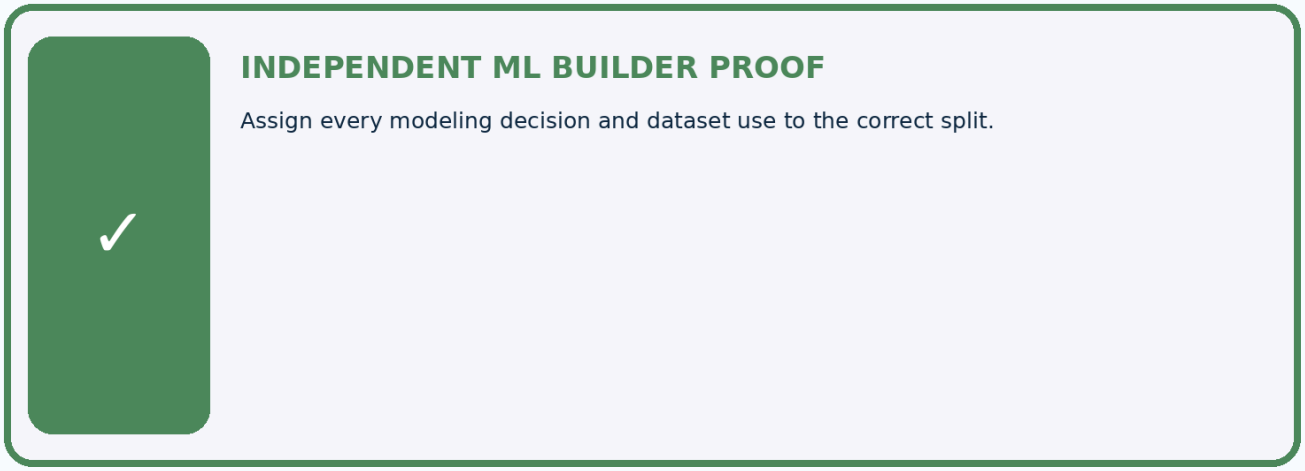
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Assign every modeling decision and dataset use to the correct split.

NEXT BRIDGE Carry the experiment **evidence** into: Metrics Match Consequences.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Split 100 independent leaves into 60 train,20 **validation**,20 test.

Explain roles and pre**process**ing.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Train fits parameters and learned transforms; **validation** chooses settings; final test evaluates the selected system. Group related records first.

Evidence or stage	Value or result
Train, fit parameters	60
Validation, choose settings	20
Test, evaluate selected system	20

▼ Earlier wording, alternatives and picture transcripts

Week 24 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Three partitions have three different jobs

Split 100 independent leaves into 60 train,20 **validation**,20 test.

Explain roles and pre**process**ing.

Train fits parameters and learned transforms; **validation** chooses settings; final test evaluates the selected system. Group related records first.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Select a model using a declared validation rule	Explain the deciding step, show a trace or test, and state one limit.
Keep training, validation and test roles separate	Explain the deciding step, show a trace or test, and state one limit.
Recognize repeated validation as development	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: Assign a separate job to **training**, **validation** and final test **evidence**.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Assign a separate job to **training**, **validation** and final test **evidence**.

- Which labelled step matches this detail? Freeze A before scoring final cases; do not select again using their labels.

- 1. Reserve final test

- 2. Fit on **training**

- 3. Select on **validation**

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Reserve final test Freeze A before scoring final cases; do not select again using their labels. Read the picture in order.

Freeze A before scoring final cases; do not select again using their labels. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. Select A using the predeclared simpler-model tie-break.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Select A using the predeclared simpler-model tie-break. The **validation** errors tie, so the stated secondary rule decides without consulting final data.

Repair and keep the record

Review Hold Out Test Data (ai-4/23). Goal: Freeze the full evaluation procedure before testing. Short practice: What is needed for an honest final claim about the revised version?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Absolute Error and MAE (ai-4/21). Goal: Calculate MAE MSE and RMSE. Short practice: Why is quoting MSE as4.67 cm misleading?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Error Analysis (ai-4/18). Goal: Categorize development errors using **evidence**. Short practice: What **evidence** would support the claim that a change helps fresh dim-light images?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

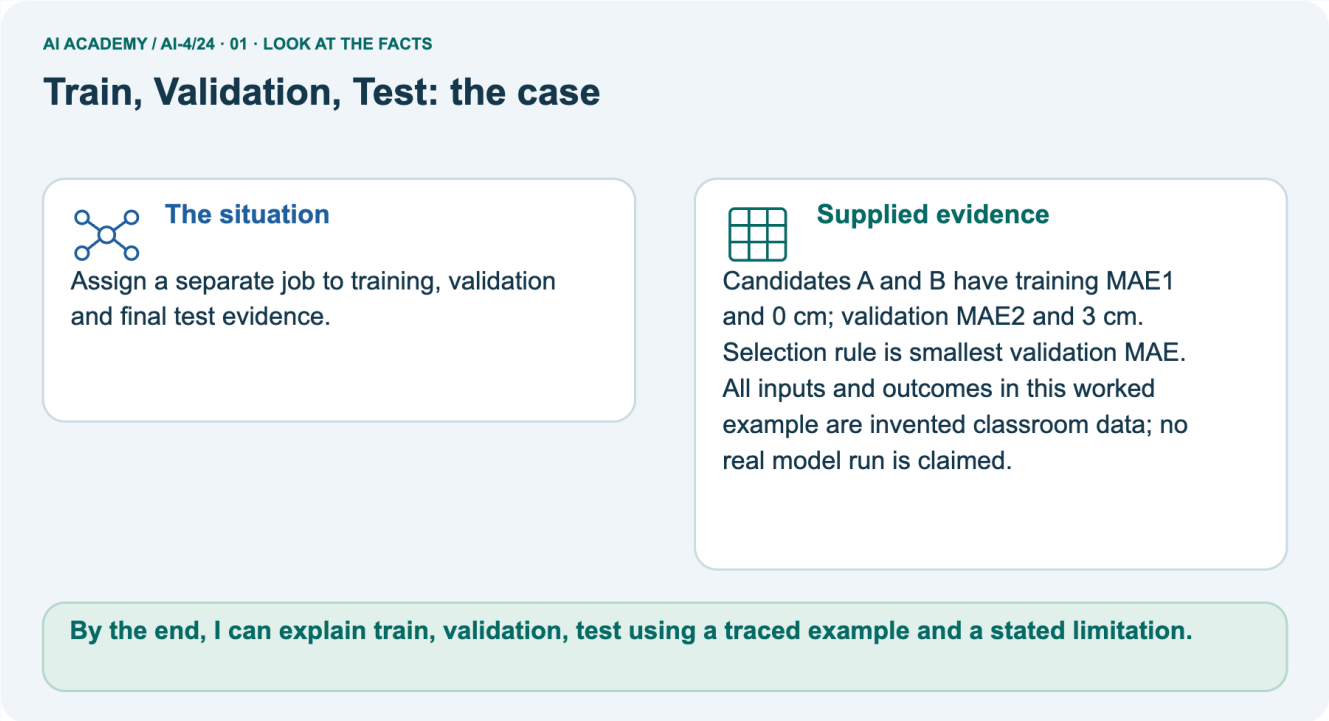
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Train, Validation, Test: the case



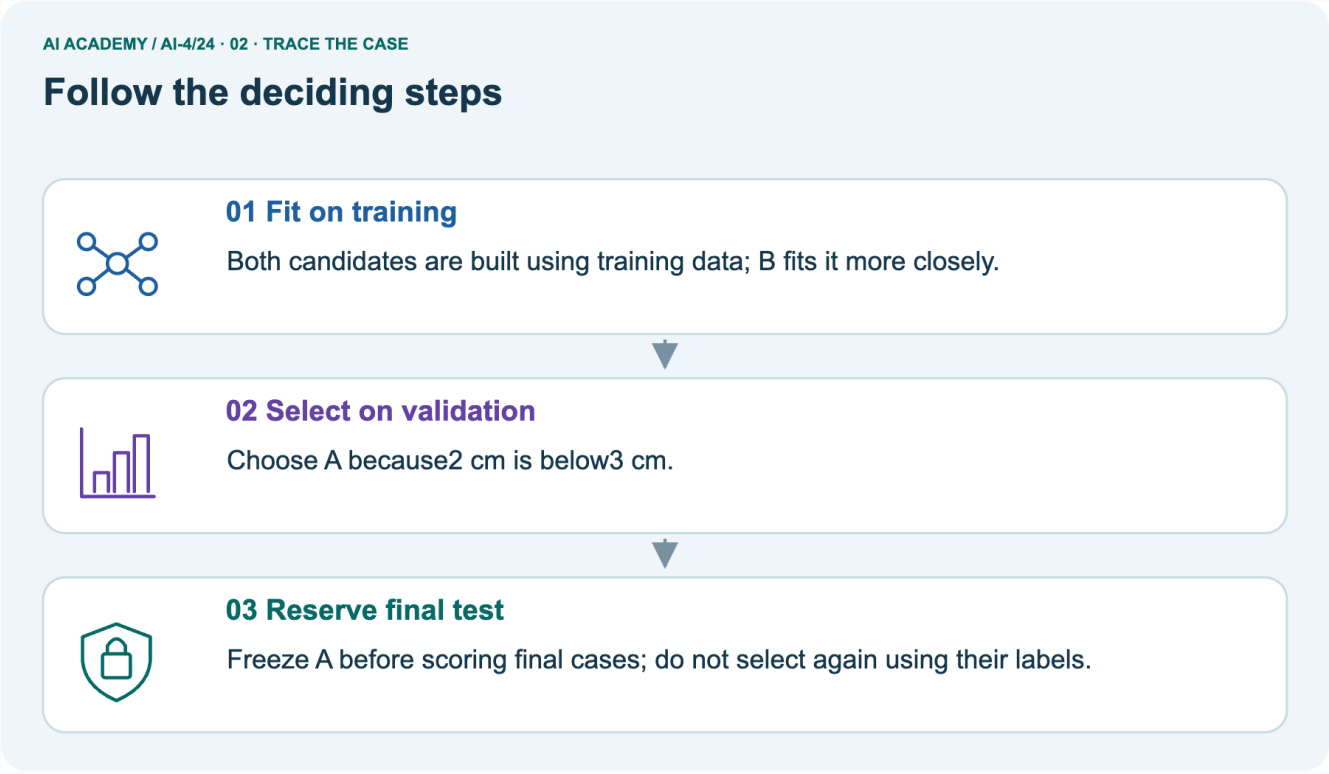
Follow the labelled picture with the matching explanation. Open the vector diagram

Assign a separate job to **training**, **validation** and final test **evidence**. Supplied facts: Candidates A and B have **training** MAE1 and 0 cm; **validation** MAE2 and 3 cm. **Selection rule** is smallest **validation** MAE.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

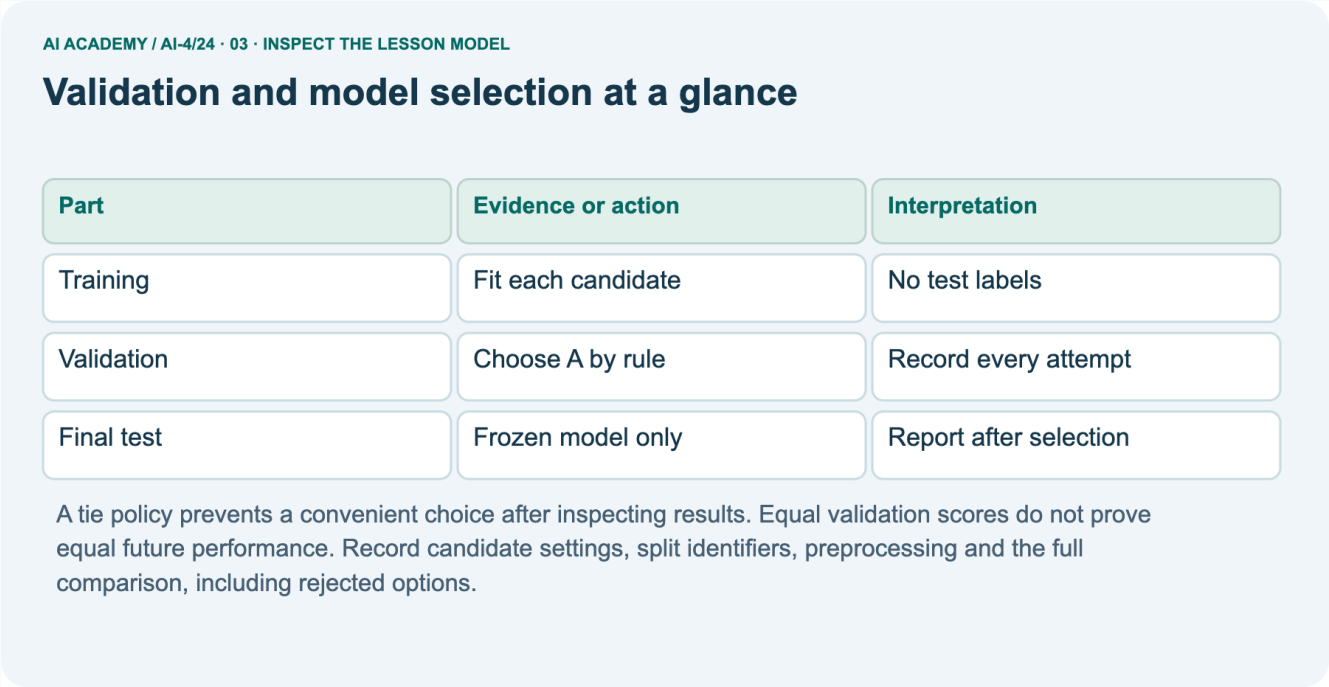


Follow the labelled picture with the matching explanation. Open the vector diagram

Fit on **training**: Both candidates are built using **training** data; B fits it more closely. → Select on **validation**: Choose A because2 cm is below3 cm. → Reserve final test: Freeze A before scoring final cases; do not select again using their labels.

Point to each numbered stage and explain which recorded **input** it uses.

03. Validation and model selection at a glance

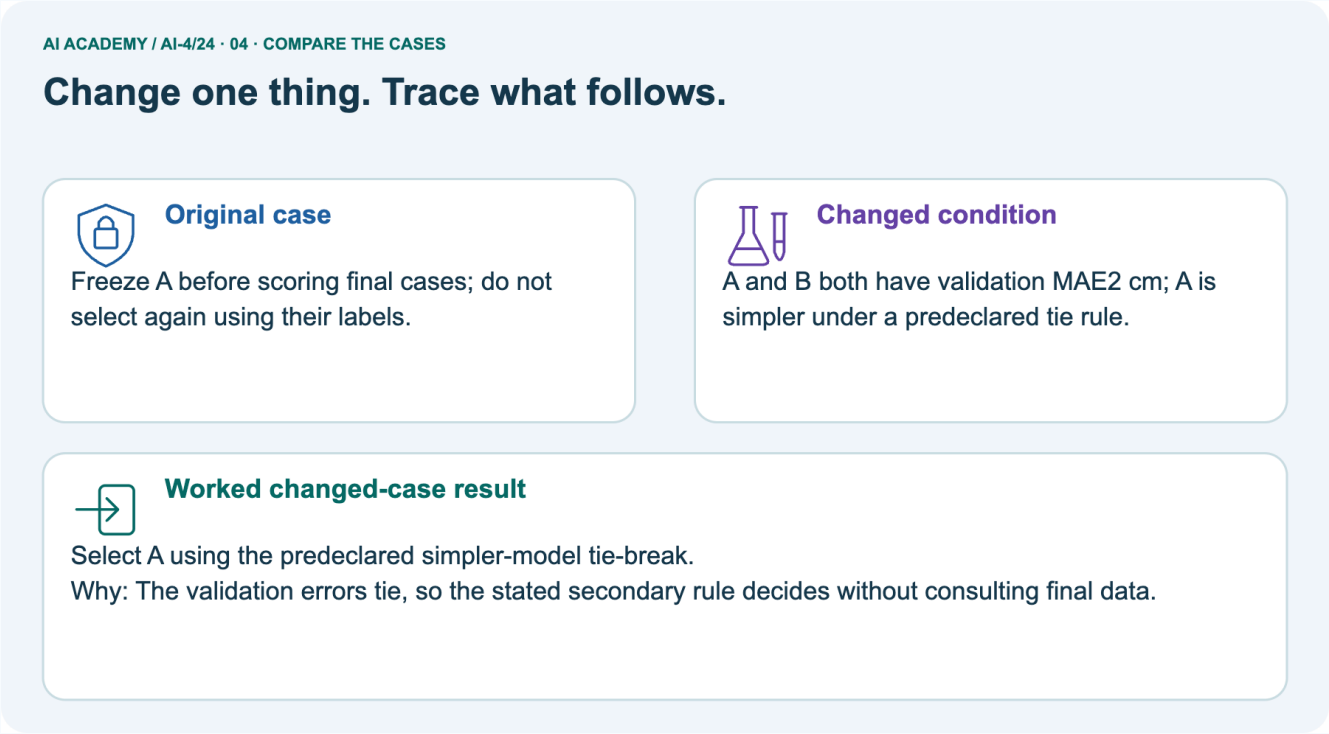


Follow the labelled picture with the matching explanation. Open the vector diagram

Validation and model selection at a glance. **Training** | Fit each candidate | No test labels. **Validation** | Choose A by rule | Record every attempt. Final test | Frozen model only | Report after selection

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



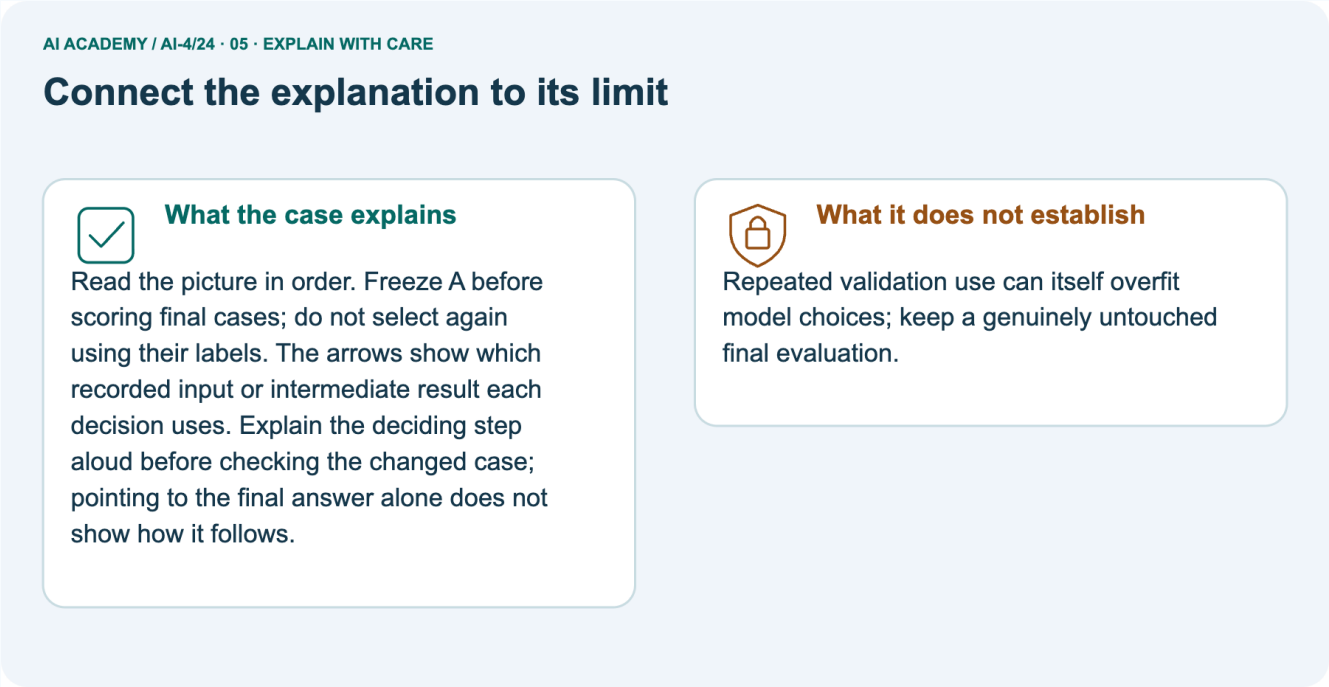
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Freeze A before scoring final cases; do not select again using their labels. Changed condition: A and B both have **validation** MAE2 cm; A is simpler under a predeclared tie rule.

Expected result: Select A using the predeclared simpler-model tie-break. Reason: The **validation** errors tie, so the stated secondary rule decides without consulting final data.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

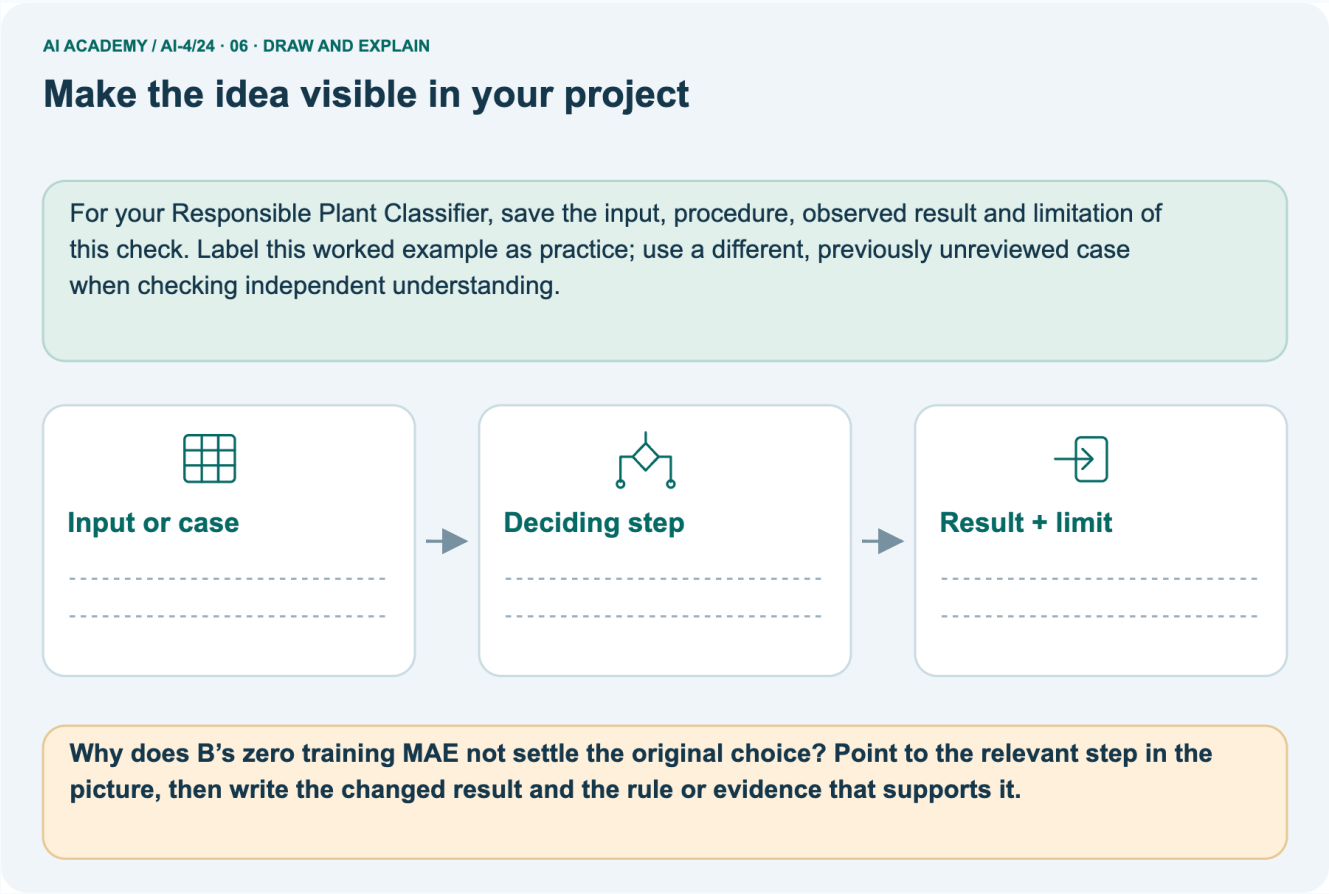
Read the picture in order. Freeze A before scoring final cases; do not select again using their labels. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Repeated **validation** use can itself overfit model choices; keep a genuinely untouched final evaluation.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why does B's zero **training** MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Train, Validation, Test: the case

02. Follow the deciding steps

03. Validation and model selection at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.